

Full T/NK Atlas Rebuild

Rebuilt from the frozen historical atlas inputs, eight independently filtered extension cohorts, and the COPD BALF/BLOOD cohort.

5,933,312 cells	27,413 genes	16 input cohorts	40 source accessions
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Scope and provenance

The historical component contains 5,128,904 cells. The eight new extension cohorts add 758,135 cells, and COPD BALF/BLOOD adds 46,273 cells. Frozen model-validation roles are retained as metadata and were not used as biological labels during integration.

TCR metadata boundary: the separately validated 14-source repaired TCR sidecar was not propagated. This rebuild preserves the source H5AD TCR fields and records that limitation explicitly.

Sample-metadata boundary: source metadata were preserved. The known GSE169246 _b blood-library compartment correction has not been written back, so tissue and specimen-context analyses involving that cohort remain blocked pending the separately reviewed additive correction.

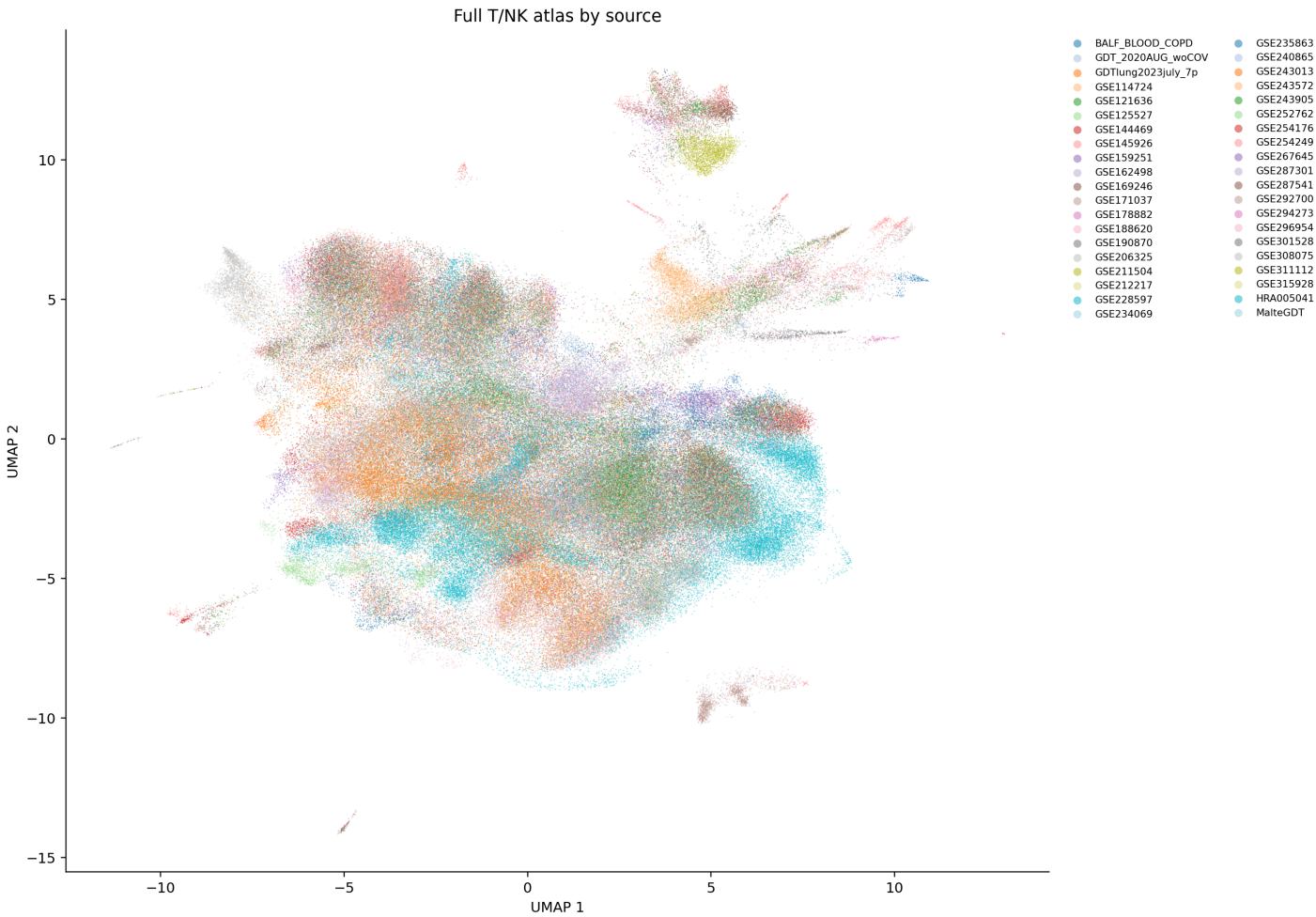
Method

All inputs were converted to sparse raw-count space, aligned to the historical 27,413-gene universe, and assigned deterministic cohort-prefixed cell IDs. The HRA005041 Seurat LogNormalize matrix was inverted using its per-cell count totals, matching the historical workflow. Four thousand source-balanced HVGs were selected after excluding TCR V/J/D, mitochondrial, ribosomal, immunoglobulin, and common noncoding features; a fixed T/NK context panel was retained. scVI used a 30-dimensional latent space on an A100 GPU, followed by RAPIDS 15-neighbor graph construction, Leiden clustering at resolution 1.0, and UMAP.

scVI runtime: 55.5 minutes. Embedding runtime: 8.4 minutes. CPU fallback: False.

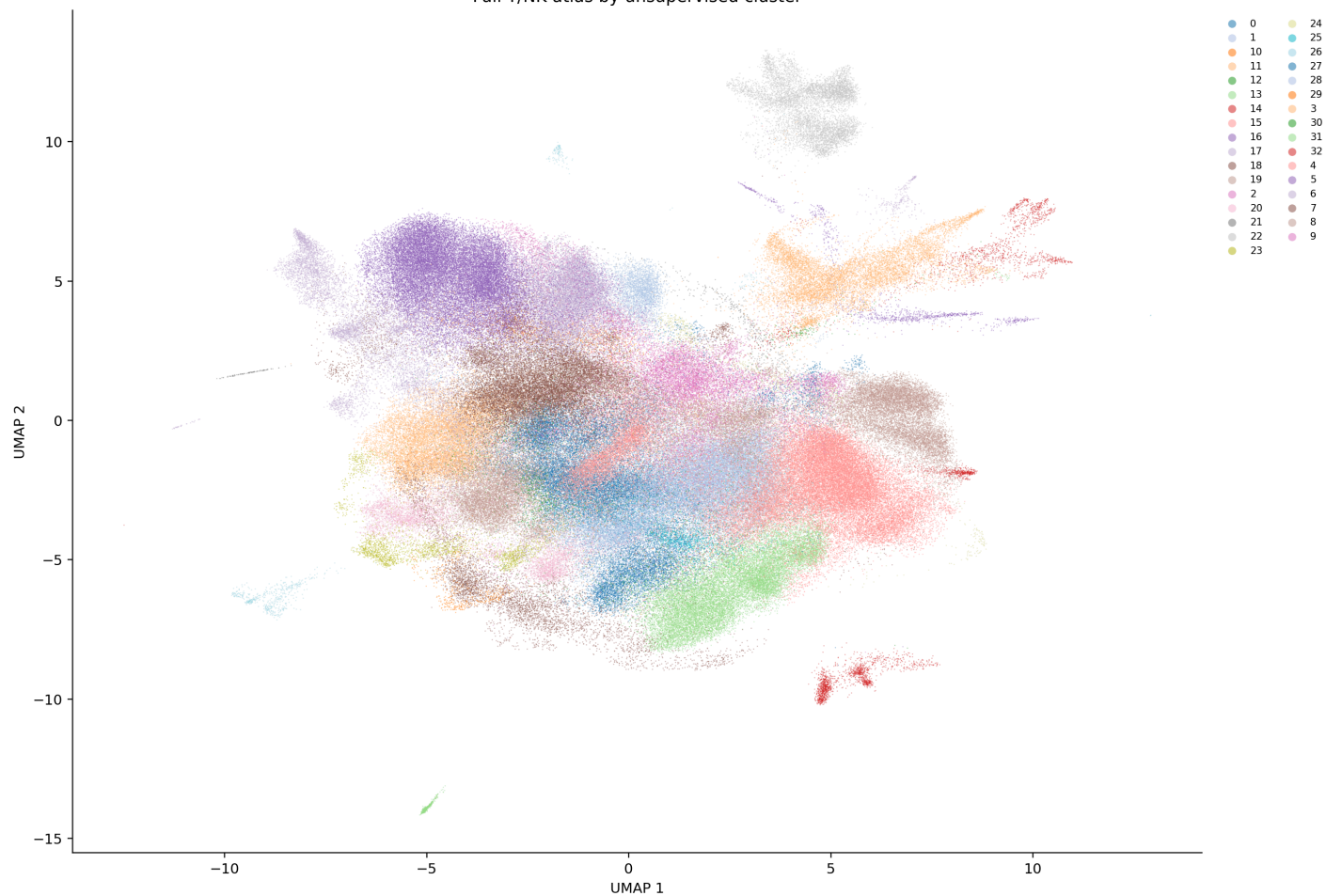
Small-batch disclosure: 2 of 1,671 integration batches contained fewer than three cells (2 of 5,933,312 atlas cells in total). These cells were retained, and the exact batches are listed in `integration_batch_counts.csv`.

Integrated structure



UMAP colored by source accession; a stratified sample is rendered for legibility.

Full T/NK atlas by unsupervised cluster



UMAP colored by unsupervised Leiden cluster.

Input composition

input_cohort_id	atlas_input_role	model_evaluation_role_frozen	n_cells
historical_core_25	historical_core	mixed_historical_roles	3705306
HRA005041	historical_expansion	legacy_gold_and_negative_truth	766639
GSE206325	historical_expansion	historical_atlas_only	501012
GSE169246	new_extension	locked_validation_cohort	354373
GSE144469	historical_expansion	legacy_gold_and_negative_truth	107068
GSE296954	new_extension	development_extension	86608
GSE292700	new_extension	development_extension	78254
GSE159251	new_extension	development_extension	68898
GSE315928	new_extension	locked_validation_cohort	66813
GSE294273_GSE294274	new_extension	development_extension	55744
BALF_BLOOD_COPD	new_extension	reused_cross_study_benchmark	46273
GSE114724	new_extension	development_extension	28652
GDT_2020AUG_woCOV	historical_expansion	gold_sorted_gdt	25904
GSE121636_GSE121637	new_extension	locked_mixed_stress_cohort	18793
GDTlung2023july_7p	historical_expansion	gold_sorted_gdt_low_quality	15175
MalteGDT	historical_expansion	gold_sorted_gdt	7800

Source composition

source_gse_id	n_cells
HRA005041	766639
GSE243013	759436
GSE206325	501012
GSE254249	406494
GSE169246	354373
GSE243905	350438
GSE287301	340227
GSE235863	241766
GSE212217	213706
GSE243572	188217
GSE228597	149812
GSE188620	123218
GSE308075	112337
GSE287541	110874
GSE144469	107068
GSE254176	96953
GSE267645	92501
GSE311112	90599
GSE296954	86608
GSE292700	78254
GSE162498	75880

source_gse_id	n_cells
GSE159251	68898
GSE178882	67755
GSE315928	66813
GSE294273	55744
GSE125527	46396
BALF_BLOOD_COPD	46273
GSE252762	45132
GSE301528	44105
GSE211504	37244
GSE171037	31937
GSE114724	28652
GDT_2020AUG_woCOV	25904
GSE240865	21680
GSE145926	21039
GSE234069	20807
GSE121636	18793
GSE190870	16753
GDTlung2023july_7p	15175
MalteGDT	7800

Tissue composition

tissue_harmonized	n_cells
	3849174
Tumor	236892
Normal	227320
tumor	177712
unresolved	118042
stomach	93885
blood	89731
breast	87879
peripheral blood	83669
jejunum	72733
lymph_node	71504
pancreas	67484
duodenum	61899
liver	58636
lymph_node_metastasis	55744
esophagus	53012
sigmoid colon	48557
chest_wall	42698
lung	37592
spleen	37330
adjacent_tissue	32315

tissue_harmonized	n_cells
PBMC	29090
breast_tumor	28652
ileum	22355
trachea	20936
mesenteric lymph node	20733
bladder	20214
heart	20039
ascending colon	19489
skin	19007
muscle	17737
BALF	17183
transverse colon	15128
lesser gastric curvature lymph node	14092
bone marrow	13741
cord blood	12498
thymus	12278
renal_tumor	9165
kidney	7514
subcutaneous_metastasis	4862
lymph node	4791

Canonical outputs

H5AD: /ssd/tnk_phase3/Integrated_dataset/full_atlas/integrated_full_atlas.h5ad
NFS link:
/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/Integrated_dataset/integrated_full_atlas.h5ad
Model: /ssd/tnk_phase3/Integrated_dataset/full_atlas/scvi_model